

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2019

Course: B. Tech. in Electronics and Telecommunication Engineering Sem: III

Subject Name and Subject Code: Digital Logic design(BTEXC305)

Max Marks: 60

Date: 01/06/2019

Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	A) Design a 4-bit binary to Gray code converter.	(3)	06
	B) Minimize the given Boolean expression using Quine-McCluskey method $F(A, B, C, D) = \sum m(0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13)$	(2,3)	06
Q. 2	A) Give comparison between Combinational and Sequential Circuit	(1,2)	06
	B) Write down Next state equation and excitation table for SR, JK, T and D flipflop.	(2)	06
Q. 3	A) Draw and explain the Serial In Parallel Out Shift Register.	(2,3)	06
	B) Design the circuit to generate the sequence : 0 --- 2 --- 5 --- 4 --- 7 --- 3.	(2,3)	06
Q. 4	A) Explain two input TTL NAND Gate	(3)	06
	B) Give comparison between TTL and CMOS logic family	(2)	06
Q. 5	A) Implement Full Adder Circuit Using PLA.	(2,3)	06
	B) Explain VHDL three modelling styles with example.	(4)	06
Q. 6	A) Design 4 : 16 decoder using 2: 4 decoder.	(4)	06
	B) Implement 3 bit Asynchronous Up Counter. Draw clock waveform.	(3)	06

***** The End *****